

Linear Regression Analysis in the Social Sciences

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Introduction to Statistics and OLS

Schedule

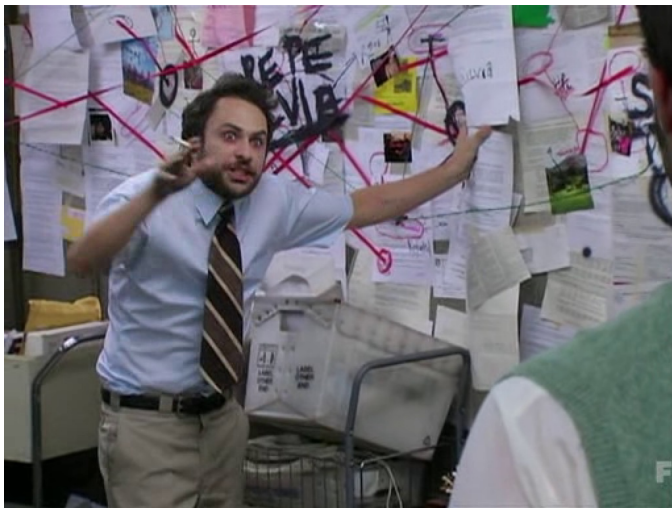
- Two sessions a day
- Each session
 - 1 Lecture (30 minutes)
 - 2 R code demonstration (20 minutes)
 - 3 Short break
 - 4 Repeat steps 1 and 2
 - 5 Lab
 - 6 Longer break

Goals

- ➊ What are **models**?
- ➋ What are **data**?
- ➌ What are **statistics**?
- ➍ Examine some examples

Social Science Research

- 1 Develop a research question
- 2 Answer question. Simple, right? Well how?



What Do We Want to Learn?

A research question implies a **target quantity** — the **estimand**.

Question	Estimand
Does education affect income?	Change in expected income per additional year of education
Does party ID predict vote choice?	Difference in vote probability across party identifiers
Does mediation reduce conflict?	Change in expected battle deaths when mediation is present

We need tools for *estimating* this quantity, starting with a model

Start simple: Simple is good

Instead of complicating things, simplify. Specify a model.



Model of PhD life: Lego grad student



*Volunteering to serve on the graduate admissions committee, the grad student is consumed by inadequacy as he skims the applicants' CVs.

A Simple Model

A student you never met asks you “What score should I expect in your class?” Lots of factors explain class performance (let’s call this y), but you want to provide a simple answer:

- Perhaps you tell every student the same answer (e.g. “42”)

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This is a model. It says:

$$\hat{y}_i = \bar{y} \quad \text{for all } i$$

- One number explains (or tries to explain) every observation.
- It’s a simplification — and it’s “wrong” for most observations.
- But it captures the center of the data: it tells us what a **typical** observation (e.g. student) will do in your course.

Finding the Center

Example: Class Exam Scores

65, 70, 70, 75, 75, 75, 80, 85, 95

- Three ways to find the "middle":
 - ▶ Mode: Most common score (75)
 - ▶ Median: Middle score when ordered (75)
 - ▶ Mean: "Share points equally" (77)

What Each Center Tells Us

- Mode (75):
 - ▶ Most frequent score
 - ▶ Shows what's "typical"
 - Median (75):
 - ▶ Half scored higher, half lower
 - ▶ Not affected by extreme scores
 - Mean (77):
 - ▶ Mathematical average
- $$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$
- ▶ Affected by all scores
 - ▶ Pulled up by high score of 95

How Typical is Typical?

If $\bar{y}$ is our model, how far off is it for each observation? In other words, how often will a student actually perform at or near the mean in your course?

$$e_i = y_i - \bar{y}$$

- These are **residuals**: the gap between what we observe for each individual student (y_i) and what our model says ($\bar{y}$).
- Some are positive (observation above the mean), some negative (below).
- A residual of zero means the model got that observation exactly right.

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The residuals show us how much **variation** exists in y

Summarizing the variation

We want a single number that captures how far off our observations are from the model.

- We can't just average the residuals — they sum to zero by construction.
- Instead, square them first, then average:

$$\text{Variance} = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n}$$

- The **standard deviation**, $s = \sqrt{\text{Variance}}$, puts this back in the original units.
- It tells us: how far is a typical observation from the mean?

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Variance is what's *left unexplained* by the simplest possible model.

Understanding Spread

How far from the center?

- Scores: 65, 70, 70, 75, 75, 75, 80, 85, 95
- Mean = 77
- Questions to consider:
 - ▶ How far is each score from 77?
 - ▶ What's a "typical" distance from 77?
 - ▶ Are scores clustered or spread out?

Standard Deviation: The Typical Distance

- Standard deviation = “typical” distance from the mean
- Formula: $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$

Standard Deviation: The Typical Distance

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- For our scores: about 8.5 points
- What this means:
 - ▶ Most scores (about 70%) are within 8.5 points of 77
 - ▶ So most scores are between 68.5 and 85.5
 - ▶ Scores outside this range are “unusual”

Why This Matters

- With mean and standard deviation, we can:
 - ▶ Identify typical values
 - ▶ Spot unusual values
 - ▶ Compare different datasets
 - ▶ Make predictions
- Example: If a new student takes the exam...
 - ▶ Score of 77 ± 8.5 would be “typical”
 - ▶ Score above 94 would be “unusually high”
 - ▶ Score below 60 would be “unusually low”

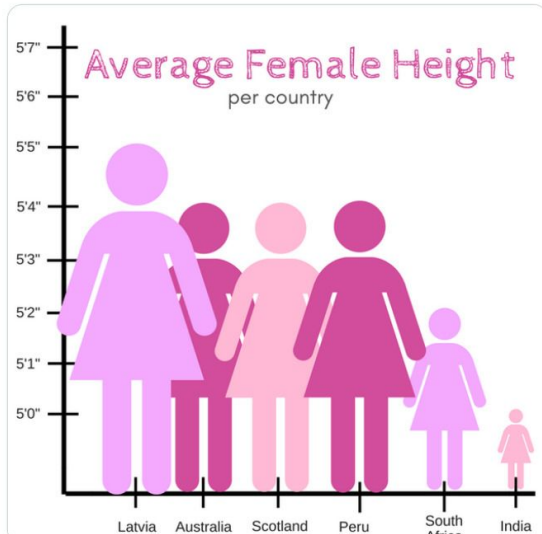


Sabah Ibrahim

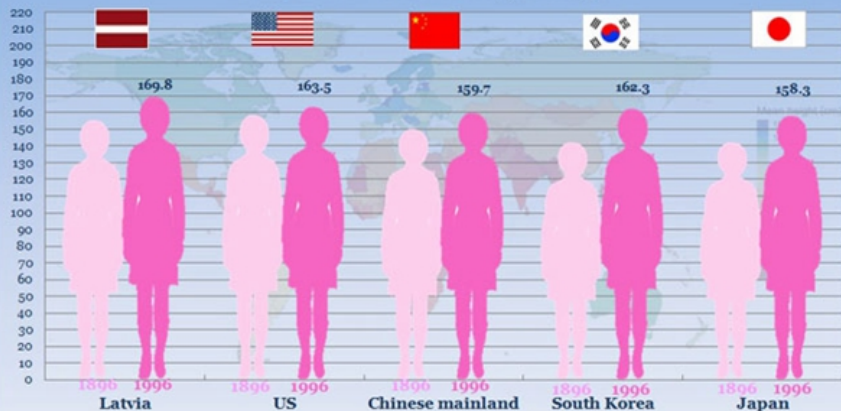
@reina_sabah



As an Indian woman, I can confirm that too much of my time is spent hiding behind a rock praying the terrifying gang of international giant ladies and their Latvian general don't find me



Comparison of Ms Average's height

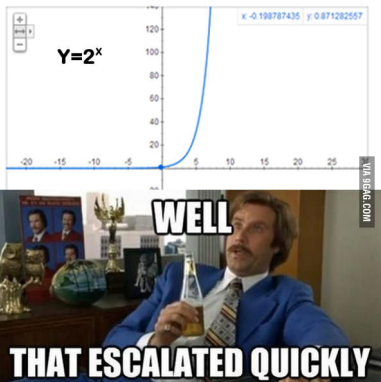


Specify a model for X and Y

- ➊ A **model**: a simplified approximation of reality
- ➋ A **statistical model**: a simplified, mathematical equation that approximates reality, usually with uncertainty.
 - ➊ $y = f(x)$
 - ➋ Life expectancy = $f(\text{Education Level}) + \varepsilon$

What's a function

- ① A **function**: relates inputs to outputs, where every input (x) produces exactly one output (y).
- ② Examples
 - ① Identify function: $y = x$
 - ② Logarithmic: $y = \ln(x)$
 - ③ Exponential: $y = 2^x$
 - ④ Linear: $y = a + bx$, where a is an intercept and b is a slope



Linear functions

- In social science research, we often rely on linear functions to say something about the relationship between 2 or more variables. Why?
 - ▶ *Easy to calculate
 - ▶ *Easy to interpret
 - ▶ Foundational to other functions and models



Linear functions

- We use:

$$(1) : y = \alpha + \beta x + \epsilon$$

instead of:

$$(2) : y = \alpha + \beta x$$

- What's the difference?
 - ▶ (1) is inferential and is probabilistic
 - ▶ (2) is descriptive and deterministic

Deterministic versus probabilistic

- Statistics assumes variation and uncertainty. What do we mean by this?
- “What is the relationship between voter choice and party ID?”
- Deterministic account of voter i 's choice :

$$vote_i = f(\text{partyID}_i).$$

- Omits all other determinants: gender, name, political knowledge, challenger quality, weather on election day, having the local college football team win the previous weekend

Probabilistic

$$vote_i = f(partyID_i) + \epsilon$$

- We live in a stochastic world:
- Other factors may affect the outcome, but are not of direct interest. We think of them as part of the natural variation in turnout.
- Error (stochastic) term represents all factors that affect vote choice. Need some way of quantifying stochastic outcomes: probability.
- Statistics are the best way to speak about the stochastic element of the model (and estimate parameters).

$$y = \alpha + \beta x + \epsilon$$

A Stochastic World

When we use statistics, we assume the data-generating process is stochastic. There are 3 sources to this uncertainty.

- Sampling error,
- Omitted variables (wrong model),
- Fundamental uncertainty (human variance).

A Stochastic World

- In social science analysis, we want to make **inferences** using some information (data!).
- We would like to make claims about the confidence of our inferences; this requires the formal use of uncertainty (or put simply, probability).

$$y_i = \alpha + \beta x_i + \epsilon_i$$

- We need to make probabilistic assumptions about $y, \alpha, \beta, \epsilon$, and x

Variables and Distributions

Variables take different values with different probabilities.

- Continuous: GDP growth, life expectancy
- Discrete: vote choice, number of conflicts

A **probability distribution** lists the possible values and the probability of each.

- $\Pr(\text{Heads}) = 0.5, \quad \Pr(\text{Tails}) = 0.5$
- $\Pr(\text{Vote}) = p, \quad \Pr(\text{Not Vote}) = 1 - p$

If using a linear model, we typically assume that y (conditional on x) follows a **normal distribution**.

Normal Distributions: Key Properties

The normal distribution is the handiest and probably the most commonly assumed distribution in all of statistics.

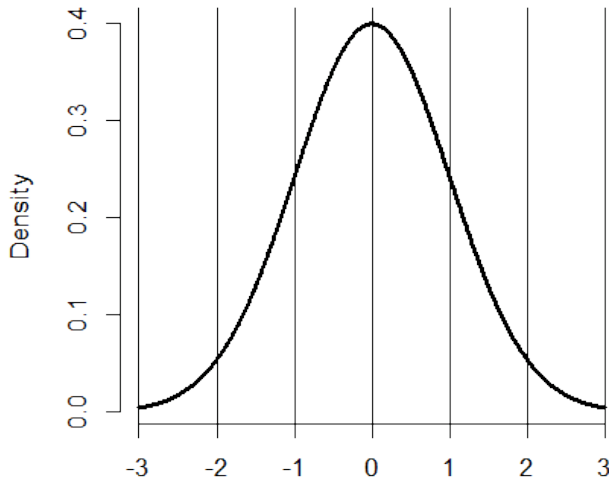
- a variable is said to follow a normal (or Gaussian) distribution with mean μ and variance σ^2 if the associated density is:

$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- μ is the mean, σ^2 is the variance
- We write $X \sim N(\mu, \sigma^2)$
- When $\mu = 0$ and $\sigma = 1$, the distribution is called a standard normal, often labeled Z . The standard normal PDF is often noted as $\phi(x)$. The standard normal CDF is denoted $\Phi(x)$.



Standard Normal Distributions



Normal Distributions

Characteristics:

- Approximately 68%, 95% and 99% of the normal density lies within 1, 2 and 3 standard deviations from the mean, respectively
- -1.28 , -1.645 , -1.96 and -2.33 are the 10^{th} , 5^{th} , 2.5^{th} and 1^{st} percentiles of the standard normal distribution respectively
- By symmetry, 1.28 , 1.645 , 1.96 and 2.33 are the 90^{th} , 95^{th} , 97.5^{th} and 99^{th} percentiles of the standard normal distribution respectively

A Note on Normality

- If you don't have a strong case for another distribution, normality is often the default assumption.
- This is not always appropriate — it matters that we use a model with the right distributional assumption.
- Normal distributions are popular partly for historical reasons: nice mathematical properties mattered when computation was expensive.
- The central limit theorem helps justify normality assumptions for sampling distributions, even when the underlying variable isn't normal.

After Theory, Before Inference

- Once you have developed a theory and collected data, don't rush into hypothesis testing.
- Know the data.
- Your theory should tell you what you expect that data to look like, but you should make sure to check.

Data

- What are **data**?
- A collection of facts, such as numbers, words, measurements, observations or even just descriptions of things.
 - ▶ **Qualitative data** is descriptive information (it describes something)
 - ▶ **Quantitative data** is numerical information (numbers)

Know thy data

- Carefully inspect your data.
- Are they categorical or continuous?
- Are the values reasonable?
- Relationship with other variables?
 - ▶ Correlations, cross-tabs, and scatter-plots
- Does your inspection of the data match your expectations of the underlying data-generating process?

Descriptive Statistics

- Next, we attempt to identify the typical observation from our variable.
- Summarize information contained in one or multiple variables.
- No attempt to make inferences to a broader population or class of events.

Beyond description: Inference

- Statistical inference is the process of drawing formal conclusions from data.
- Use the facts we know to learn about stuff we don't know.
- Unknown? : true relationship between x and y
- Known? Data

Questions to ask when you have your data

Are Your Data Explainable by Chance Alone?

- You toss a coin 10 times, and it turns up heads only twice.
- Is the difference between 20% heads and the expected 50% heads result explainable by chance alone?
- The phrase “the results are explainable by chance alone” means that the difference you observe in your data is within the expected range of results that you expect when there is no difference in reality.

Are Your Data Explainable by Chance Alone?

To answer this question, you will want to develop a theoretical model of no difference: a null model.

- A null model would state there is no difference in the process that produced these data.
- A test statistic, which is in general a function of your sampled data, is used to test this model. The subject of statistical significance testing, also called hypothesis testing, gives you an answer to the question, “Are the results explainable by chance alone?”
- If chance alone explains how the data arise, then there is no difference between the distributions of any of the observations.

APPENDIX

A Stochastic World

The implications of our stochastic DGP?

- Data are noisy
- Data are imperfect reflections of the underlying DGP
- No dataset is the realization of the entire social process of interest
- Any measure of the data will take on this noise and randomness.

Distributions

- Distributions allow us to describe (and then analyze) the set of values that variables may take
- Some distributions appear so frequently (or are often assumed to characterize the population) that we give them names. Let's start with Bernoulli.
- A variable Y that takes on only two values (0,1) with $P(Y = 1) = p$ have a Bernoulli distribution with parameter p .

$$P(Y = y \mid p) = \begin{cases} 1 - p & \text{for } y = 0 \\ p & \text{for } y = 1 \end{cases}$$

Can be written as...

$$P(Y = y \mid p) = (p)^y(1 - p)^{1-y} \quad (1)$$

The Bernoulli Distribution

- The Bernoulli distribution describes randomly produced binary variables where the outcomes are independent, i.e. flipping coins.
- Political examples include the expected frequency of valid vs. spoiled ballots; conflict, alliances, signing human rights treaties.
- The Bernoulli distribution underpins several other types distributions (i.e. binomial).

Quantiles

You've heard of sample quantiles.

- If you were the 95th percentile on an exam, you know that 95% of people scored worse than you and 5% scored better.
- You also heard of a median. The median is the 50% percentile.
- The α^{th} **quantile** of a distribution with distribution function F is the point x_α so that

$$F(x_\alpha) = \alpha$$

- A **percentile** is simply a quantile with α expressed as a percent

Quantile Example

The 95th percentile of a distribution is the point so that:

- The 0.95 quantile of a distribution is the point so that 95% of the mass of the density lies below it. Or, in other words, the point so that the probability of getting a randomly sampled point below it is 0.95.
- the probability that a random variable drawn from the population at this point or less is 95%
- the probability that a random variable drawn at a higher point is 5%

Descriptive Stats

- Descriptive statistics is a useful way to communicate patterns within or between data.
- Given space constraints, journal articles usually don't show much in the way of descriptive stats. That doesn't mean that they are not important.

Descriptive Stats

Let's briefly describe ways to know the data.

- What is the unit of analysis/observation.
- What cases does the data cover?
- What is the temporal scope of the data?

Console

Terminal x

Jobs x

/cloud/project/ ↗

> mtcars

	mpg	cyl	displacement	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.40	0	0	3	3
Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	0	3	3
Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.00	0	0	3	3
Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.98	0	0	3	4
Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.82	0	0	3	4
Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	0	3	4
Fiat 128	32.4	4	78.7	66	4.08	2.200	19.47	1	1	4	1
Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	1	4	1
Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	0	3	1
Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.87	0	0	3	2
AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.30	0	0	3	2
Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.41	0	0	3	4
Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.05	0	0	3	2
Fiat X1-9	27.3	4	79.0	66	4.08	1.935	18.90	1	1	4	1
Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	1	5	2
Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	1	5	2
Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	1	5	4
Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6
Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	1	5	8
Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	1	4	2

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Central Tendency and Variation

- How are observations distributed across the range of values they could potentially take on?
- What's the center of the distribution? How much variation is there around the center?
- Can we visualize it or summarize it one or a handful of numbers?

Central Tendency

Modes, Medians, Means

- Mode — most common value in the distribution.
- Median — midpoint of distribution; half of observations are greater, half are less.
- Mean — center of gravity of distribution, $\bar{x} = \frac{\sum x_i}{N}$

Central Tendency

The Mean

- Does not have to be located at center of distribution.
- Does not need to be anywhere close to the most common value.
- Values below mean must balance values above the mean.

Measures of Variability

The Variance

- Want a measure that accounts for every observation.
- What about deviations from the mean? $X_i - \bar{X}$
- What if we sum them? $\Sigma(X_i - \bar{X})$

Measures of Variability

The Variance

- If we sum them, they may cancel each other out.
- What if we sum the squared deviations instead? $\sum (X_i - \bar{X})^2$
- Divide by number of observations, N . $\frac{\sum (X_i - \bar{X})^2}{N}$

Measures of Variability

The Standard Deviation

- Variances lead to another problem because we square everything. They are on a different scale from original variable.
- Use the standard deviation! $\sqrt{\frac{\sum (X_i - \bar{X})^2}{N}}$

Measures of Variability

The Standard Deviation

- What is the standard deviation intuitively.
- Typical amount of change within a particular variable.
- How many standard deviations does an observation lie above or below the mean.

Central Tendency

Advantages and Disadvantages

- Mode is only measure of central tendency for nominal variables.
- Median can be “jumpy”, but not greatly affected by adding outliers.
- Mean is greatly influenced by outliers, but moves incrementally.

Central Tendency

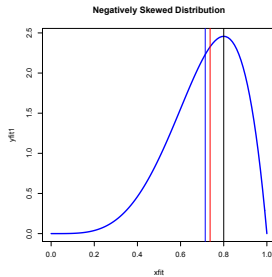
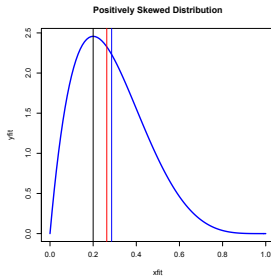
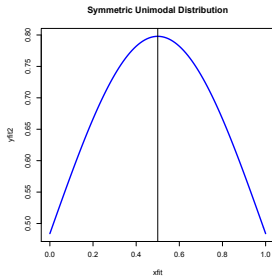
Advantages and Disadvantages

Var 1	Var 2
1	1
2	2
4	4
6	6
7	7
8	8
10	10
12	12
13	1050

Same median, different mean!

Central Tendency

Comparisons



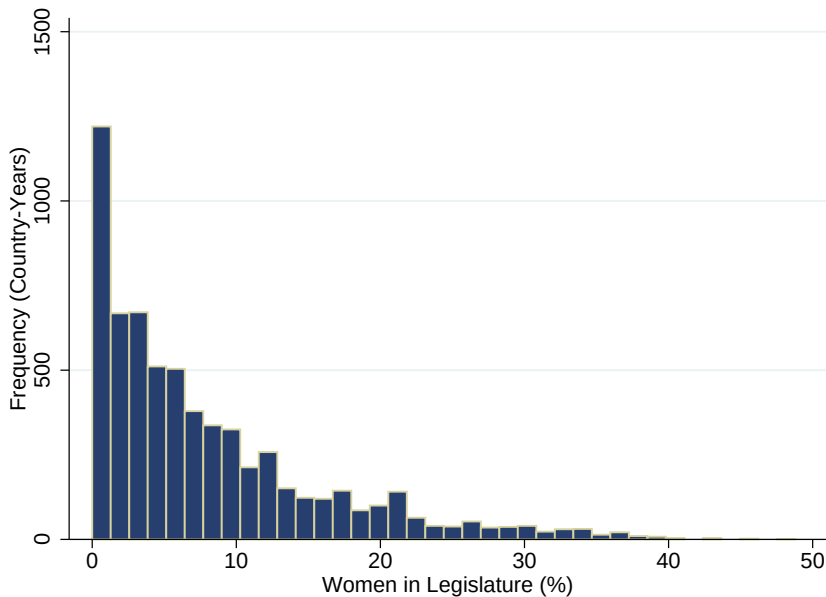
Frequency Distributions

- How can we visually summarize information contained in a single variable?
- Frequency distributions tell us each unique value a particular variable takes on and how often each value occurs
- Cumulative frequency distributions tells us the total number of cases within a particular category and all the categories below it.

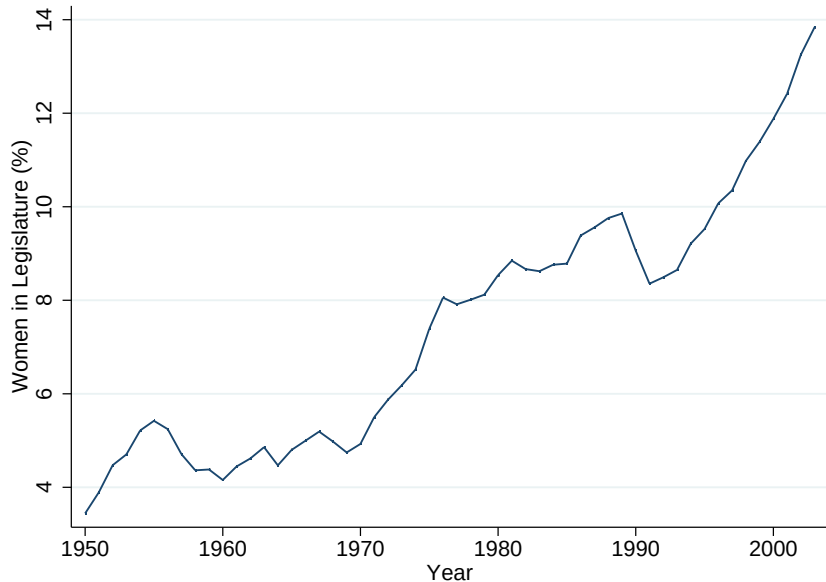
Frequency Distributions

Category	Frequency	Cumulative Frequency	Proportion	Cumulative Proportion
57-59	1	1	0.03	0.03
60-62	5	6	0.15	0.18
63-65	3	9	0.09	0.27
66-68	12	21	0.36	0.64
69-71	6	27	0.18	0.82
72-74	6	33	0.18	1.00

Histogram



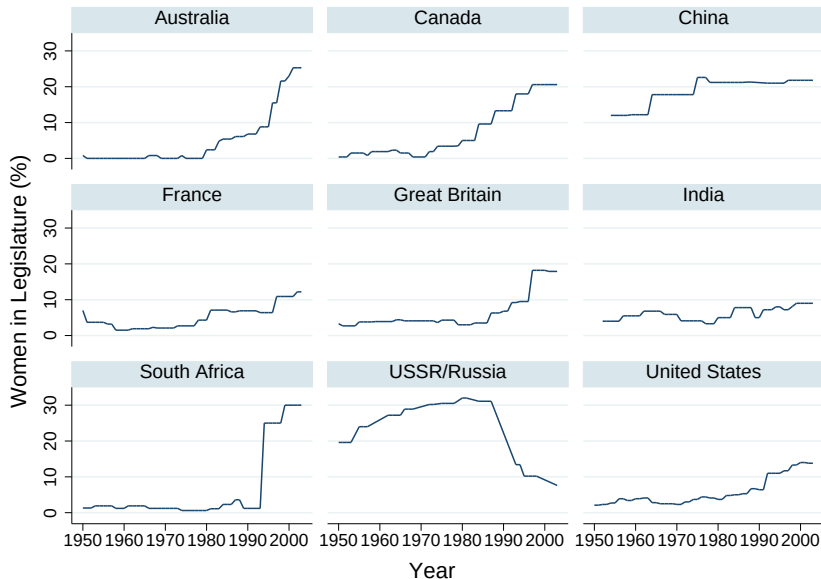
Time Series Line



Box Plots

- Plot median, IQ range, fences, and outliers
- Fences are $1.5 * IQR$ or most extreme value beyond IQR.

Time Series by Group



Types of Patterns

- Correlation
 - ▶ Two variables move together linearly.
- Causation
 - ▶ Change in an independent variable leads to change in a dependent variable.
- But how do we know which variable is dependent and which is independent?
- How do we know if any relationship we find is causal?

Covariance and Correlation

We are interested in our concepts relate to each other, thus we are interested in how variables co-vary with each other.

- Correlations and covariance are ways to measure if variables' have dependence
- But they only capture linear dependence

Covariation

Let X and Y be two RV's with means μ_X and μ_Y , respectively. Covariance is the expected value of the products of deviations from the means

$$COV(X, Y) = \sigma_{XY} = E[(X - E[X])(Y - E[Y])]$$

Covariance

The covariance between two RV's (X, Y) measures the degree to which X and Y tend to be large at the same time or the degree to which one tends to be large while the other is small.

- Suppose $\text{Cov}(X, Y) > 0$.
- Then we know that $X > \mu_X$ and $Y > \mu_Y$ and/or $X < \mu_X$ and $Y < \mu_Y$ must occur together at a larger extent than $X < \mu_X$ occurs with $Y > \mu_Y$ and $X > \mu_X$ occurs with $Y < \mu_Y$
- What about $\text{Cov}(X, Y) < 0$, $\text{Cov}(X, Y) = 0$?

Correlation

Correlation is just scale free covariance. Covariation can be affected by arbitrary scaling of a variable (i.e. $\text{Cov}(2X, Y) = 2\text{Cov}(X, Y)$.)

$$\rho_{X,Y} = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y},$$

Linear Association

- Covariance and Correlation are measures of the degree of linear association.
- Independent RV's have correlation 0, but correlation 0 does not guarantee independence!
- For example, suppose we have a sample of X 's, where X takes on the value -1, 0, and 1 with equal probability. $f(X) = y = x^2$
- We would find the X and Y have low correlation, but Y is obviously dependent on X .
- This is a limitation of statistical strategies that rely on linear techniques such as correlation and covariance; it only focuses on one type of relationship between X and Y , a linear relationship.

Motivating example: who's going to win the election?

- Unknown (estimand): who will vote for each candidate.
- We cannot poll everyone.
- What can we do?

Inference

- We make inferences about a population based upon a random sample.
- Estimate parameters values based upon statistics calculated from random samples.

Populations and Samples

- Population is the universe of all possible observations.
- Population is unobservable, theoretical concept.
- Only get to view a single (hopefully random) sample.

The election example illustrates many of the difficulties of trying to use data to create general conclusions about a population.

- Is the sample representative of the population that we'd like to draw inferences about?
- Are there known and observed, known and unobserved or unknown and unobserved variables that contaminate our conclusions?
- Is there systematic bias created by missing data or the design or conduct of the study?

US Presidential Election predictions

- Alan Lichtman (History at American U.) has predicted the winner of every presidential election all 9 elections since 1984 except one (2000).
- Doesn't use any polls, just 13 true/false questions.
- He predicted Trump's win in 2016.
- Does he have predictive value? Does he do better than random guessing?

Are Your Data Explainable by Chance Alone?

To answer this question, you will want to develop a theoretical model of no difference: a null model.

- A null model would state there is no difference in the process that produced these data.
- A test statistic, which is in general a function of your sampled data, is used to test this model. The subject of statistical significance testing, also called hypothesis testing, gives you an answer to the question, “Are the results explainable by chance alone?”
- If chance alone explains how the data arise, then there is no difference between the distributions of any of the observations.

Extended example

Suppose I collect age data from a class of undergraduates and notice that the average age of those seated in the back rows of the class is 27.5 years, while the average age of those seated in the front rows is 26.125 years.

- I am tempted to conclude that older students tend to sit in the back rows. But is the difference explainable by chance alone?
- The difference in the data is $27.5 - 26.125 = 1.375$ years. This statistic is an example of a test statistic
- What does this tell us? Not much by itself.

Extended example

Let's proceed with a randomization method to develop our null model.

- This process does not depend on unknown parameters (a theoretical GDP); The expected differences are completely known.
- Here's how it works. Students with ages 36, 23, 22, 27, 26, 24, 28, 23, 30, 25, 22, 26, 22, 35, 24, and 36 wander into class the first day. The first student in selects a seat at random and sits down. The second student in selects a seat at random from the seats still available and so on.
- It is clear that in this model there is no difference in the ages of people sitting in the front versus back seats, since the selections are random.
- This is also called a permutation model: all possible permutations of the data values are equally likely.

Extended example

Randomization method.

- Each permutation, or shuffle, gives averages for the first eight (in the front rows) and the last eight (in the back rows), and the difference between these two is purely explained by chance, because, again, there is truly no difference between front and back seats in this model.

Randomization Scenario	Student Ages, from Front to Back of Class	Average Age in the Front of Class ($n_1 = 8$)	Average Age in the Back of Class ($n_2 = 8$)	Difference between Averages, Back Minus Front
1	27 24 22 23 24 22 36 26 36 28 23 26 25 30 35 22	25.5	28.125	2.625
2	24 25 23 28 35 36 23 26 36 30 27 24 26 22 22 22	27.5	26.125	-1.375
3	35 24 30 26 36 36 22 23 24 26 25 22 22 27 23 28	29.0	24.625	-4.375
4	22 23 36 30 24 26 24 22 23 27 25 36 22 28 26 35	25.875	27.75	1.875
5	25 28 26 24 23 30 36 22 27 36 26 22 24 23 35 22	26.75	26.875	0.125
6	36 23 30 24 22 28 36 26 26 22 23 35 27 25 24 22	28.125	25.5	-2.625
7	22 25 30 36 36 22 24 23 27 26 28 26 35 23 22 24	27.25	26.375	-0.875
8	22 25 26 22 36 36 24 27 35 23 22 28 24 26 23 30	27.25	26.375	-0.875
9	28 30 22 24 36 23 35 23 27 26 22 26 25 22 24 36	27.625	26.0	-1.625
10	26 23 24 22 23 22 27 28 35 26 30 22 25 24 36 36	24.375	29.25	4.875

Extended example

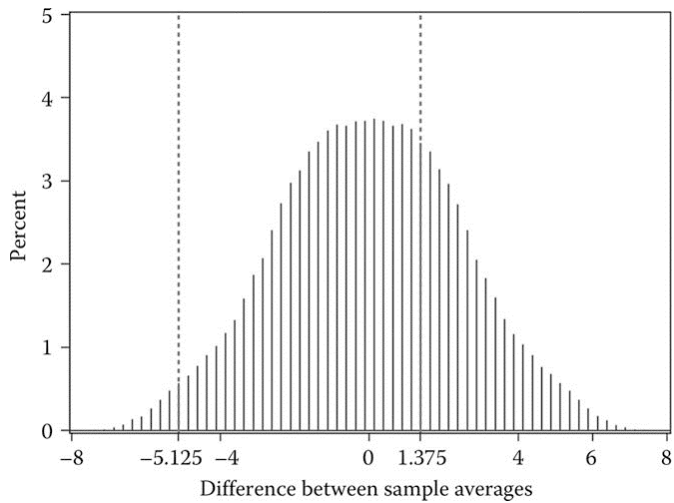
The numbers in the final column are purely chance differences.

- Further, the actual average age difference you saw from the actual students, 1.375 years, appears well within the range of these differences shown.
- Thus, the difference you see in your data, 1.375, is easily **explainable** by chance alone.
- But you cannot say that your observed difference, 1.375, **is explained** by chance alone. In other words, you have not proven that seat selection is independent of age in general.

Extended example

What if the difference were 5.125 years? Would that difference be explainable by chance alone?

- The difference 5.125 seems outside the range of chance difference, but that's only from observing 10 possible seat selections.
- Simulate and graph this process.



Extended example

The distribution shown is called the null distribution of the test statistic.

- The observed difference, 1.375 years, is easily explainable by chance alone, since it is well within the range of the permutation distribution.
- On the other hand, if the observed difference were -5.125 years then chance is much less likely as an explanation, given its placement toward the tail of the distribution under the permutation model where seat selection is truly independent of age.

Defining Sampling Distributions

A critical part of inferential statistics involves determining how far sample statistics are likely to vary from each other and from the population parameter. These determinations are based on sampling distributions.

- A sampling distribution can be thought of as a relative frequency distribution with a very large number of samples.
- More precisely, a relative frequency distribution approaches the sampling distribution as the number of samples approaches infinity.

The Central Limit Theorem

- Instead of randomizing our data as in the previous example, we can use the Central Limit Theorem to give us an idea of what the underlying sampling distribution is of estimates of interest.
- The **Central Limit Theorem** (CLT) is one of the most important theorems in statistics
- For our purposes, the CLT states that the distribution of averages of iid variables (properly normalized) becomes that of a standard normal as the sample size increases
- The CLT applies in an endless variety of settings

The Central Limit Theorem

- The result is that

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\text{Estimate} - \text{Mean of estimate}}{\text{Std. Err. of estimate}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$$

has a distribution like that of a standard normal for large n .

- (Replacing the standard error by its estimated value doesn't change the CLT)
- The useful way to think about the CLT is that $\bar{X}_n$ is approximately $N(\mu, \sigma^2/n)$

Example

- Simulate a standard normal random variable by rolling n six sided dice.
- Let X_i be the outcome for die i
- Then note that $\mu = E[X_i] = 3.5$
- $\text{Var}(X_i) = 2.92$
- $\text{SE } \sqrt{2.92/n} = 1.71/\sqrt{n}$
- Let's roll n dice, take their mean, subtract off 3.5, and divide by $1.71/\sqrt{n}$ and repeat this over and over

Extended example

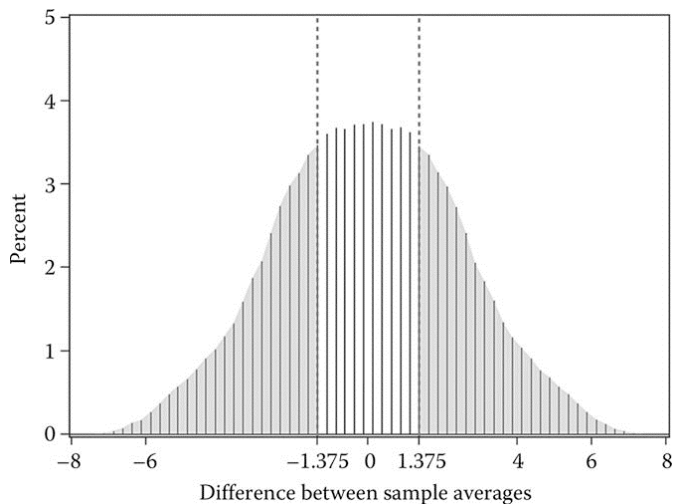
Continuing with this example, let's suppose another null model, this time using iid assumptions.

- Here our test statistic is $T = \bar{Y}_2 - \bar{Y}_1$
- We know from CLT that $\bar{Y}_2$ and $\bar{Y}_1$ are normally distributed and thus T is normally distributed.
- The null distribution of the test statistic T is approximately the normal distribution with mean 0 and standard error of $\frac{\sigma}{4} (\sigma[\frac{1}{8} + \frac{1}{8}])$
- We don't know σ , but we know that the sample sd is a consistent estimate of σ .

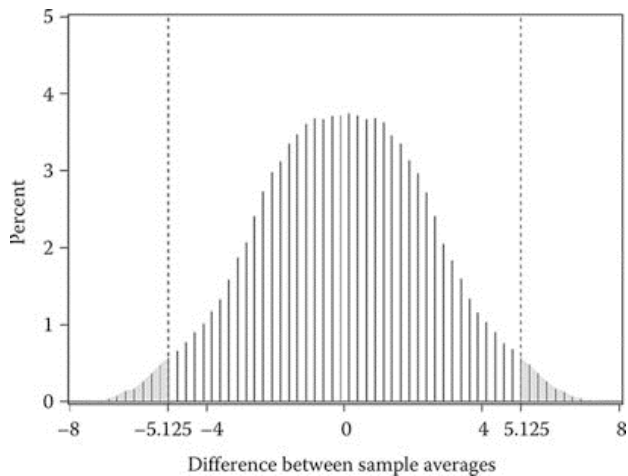
Extended example

- With some rounding error, our standard error is 2.393.
- using the normal approximation, the ± 1 , ± 2 , and ± 3 standard error ranges for the difference $\bar{Y}_2 - \bar{Y}_1$ are $(-2.393, 2.393)$, $(-4.786, 4.786)$, and $(-7.179, 7.179)$, respectively, appearing to capture roughly 68%, 95%, and 99% of the probabilities.

p-value is shaded area (0.63)



p-value is shaded area (0.043)



P-value

- Be careful that you do not say “there is a 0.043 probability that the mechanism is chance alone.”
- That interpretation is wrong, again because it conflates $\Pr(\text{chance mechanism} \mid \text{difference of } 5.125)$ with $\Pr(\text{difference is as extreme as } -5.125 \mid \text{chance mechanism})$. The latter probability is equal to the p-value, not the former.
- Remember that we assume that the null is true (so we don't assign any probability that it is true).
- Given that the null is true, a p-value is the probability that our random test statistic is at least as extreme (relative to the expected value of the sampling distribution of the null) as the observed value. The provide information of the degree of support for the hypotheses.

Null distribution

- Note that we have a good idea what the null distribution is before we look at the data. Why? Because we know the characteristics of our test distributions, and thus know how far we have to be from the mean to be considered unlikely.
- Decision rules - how you draw the line between values that are typical, and therefore explainable by chance, and values that are so unusual that chance alone can be ruled out - are also made before the data.
- The p-value is the probability of seeing a difference as extreme or more extreme than the difference that you observed, assuming your data come from a process where there is, in reality, no difference.
- The level of significance of your statistical procedure is the p-value threshold that you use to categorize a result as either explainable by chance alone or not explainable by chance alone.

Null distribution

- The normal model is never precisely correct in practice (due to discreteness, skewness, and other non-normal aspects of your data), but the normal model provides a useful anchor.

Assumptions of Normality

- What can go wrong here is that the estimate of σ can be very inaccurate with small n .
- In practice, you have no way of knowing whether your estimated σ is too small.
- You can correct for the problem by properly accounting for the variability in your estimate of σ ; this is the purpose of using Student's t-distribution rather than the standard normal distribution.

Student's t

- Student's t-distribution gives you a larger critical value than 1.96, one that explicitly accounts for the variability in your estimate of σ and will make your interval for μ exactly 95% when your data are produced as iid from a normal distribution.
- (Note: Student was William Gossett, who worked as a quality control technician at the Guinness brewery .)
- The variability of $\hat{\sigma}$ depends on another distribution, the Chi-Squared Distribution